KENT PEARCE **MATH 1351 Spring 2004**

Homework Set HW3 due 2/9/04 at 12:00 PM

You may need to give 4 or 5 significant digits for some (floating point) numerical answers in order to have them accepted by the computer.

1.(1 pt) Problem 15, Section 2.3 page 78.

The given function is defined for $x > 0$ except for $x = 2$. Find the value to be assigned to $f(2)$, if any, to guarantee that f is continuous at 2.

$$f(x) = \frac{x^2 - x - 2}{x - 2}$$

$f(2) =$ ______

2.(1 pt) Problem 16, Section 2.3 page 78.

The given function is defined for $x > 0$ except for $x = 2$. Find the value to be assigned to $f(2)$, if any, to guarantee that f is continuous at 2.

$$f(x) = \sqrt{\frac{x^2 - 4}{x - 2}}$$

$f(2) =$ ______

3.(1 pt) Problem 18, Section 2.3 page 78.

The given function is defined for $x > 0$ except for $x = 2$. Find the value to be assigned to $f(2)$, if any, to guarantee that f is continuous at 2.

$$f(x) = \frac{\cos(\pi/x)}{x - 2}$$

$f(2) =$ ______

4.(1 pt) Problem 22, Section 2.3 page 78.

Determine whether the given function is continuous (C) or not continuous (NC) on the given interval. Note: your answer should be **C** or **NC** .

$$f(x) = \begin{cases} x^2 & \text{if } 0 \le x < 2 \\ 3x + 1 & \text{if } 2 \le x < 5 \end{cases}$$

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5.(1 pt) Problem 23, Section 2.3 page 78.

Determine whether the given function is continuous (C) or not continuous (NC) on the given interval. Note: your answer should be **C** or **NC** .

$$g(t) = \begin{cases} 15 - t^2 & \text{if } -3 < t \le 0 \\ 2t & \text{if } 0 < t \le 1 \end{cases}$$

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6.(1 pt) Problem 24, Section 2.3 page 78.

Determine whether the given function is continuous (C) or not continuous (NC) on the given interval. Note: your answer should be **C** or **NC** .

$$f(x) = x\sin(x) \text{ on } (0, \pi)$$

______

7.(1 pt) Problem 39, Section 2.3 page 79.

For what value of the constants a and b is the function f continuous for all x

$$f(x) = \begin{cases} (ax + 3) & \text{if } x > 5 \\ 8 & \text{if } x = 5 \\ x^2 + bx + 1 & \text{if } x < 5 \end{cases}$$

$a =$ ______

$b =$ ______

8.(1 pt) Problem 43, Section 2.3 page 79.

For what value of the constants a and b is the function f continuous for all x

$$f(x) = \begin{cases} \dfrac{\sin(ax)}{x} & \text{if } x < 0 \\ 5 & \text{if } x = 0 \\ x + b & \text{if } x > 0 \end{cases}$$

$a =$ ______

$b =$ ______

9.(1 pt) Problem 40, Section 2.3 page 79.

For what value of the constants a and b is the function f continuous for all x

$$f(x) = \begin{cases} \frac{(ax-4)}{(x-2)} & \text{if } x \ne 2 \\ b & \text{if } x = 2 \end{cases}$$

$a =$ ______

$b =$ ______

10.(1 pt) These are exercises 13 through 16 on Section 2.4 on page 89.

Evaluate the following expressions.

(a) $\log_2(4) + \log_3\left(\frac{1}{9}\right) =$ ______

(b) $2^{\log_2(3) - \log_2(5)} =$ ______

(c) $5\log_3(9) - 2\log_2(16) =$ ______

(d) $\left(\log_2\left(\frac{1}{8}\right)\right)(\log_3(27)) =$ ______

11.(1 pt) These are exercises 23 through 26 on Section 2.4 on page 89.

Solve the logarithmic and exponential equations.

(a) $\log_x(16) = 2$

$x =$ ______

(b) $\log(x) = 5.1$

$x =$ ______

(c) $e^{-3x} = 0.5$

$x =$ ______

(d) $\ln(x^2) = 9$

$x =$ ______

12.(1 pt) These are exercises 31, 32 and 37, 38 on Section 2.4 on page 89.

Solve the logarithmic and exponential equations.

(a) $3^{x^2-x}=9$ has two solutions whose sum is ____
(b) $4^{x^2+x}=16$ has two solutions whose sum is ____
(c) $e^{2x+3}=1$
$x=$ ____
(d) $\dfrac{e^{x^2}}{e^{x+6}}=1$ has two solutions whose sum is ____

13.(1 pt) These are exercises 39 and 40 on Section 2.4 on page 89.

Solve the logarithmic and exponential equations.

(a) $\log_3(x)+\log_3(2x+1)=1$ has two solutions whose sum is ____

(b) $\ln\left(\dfrac{x^2}{1-x}\right)=\ln(x)+\ln\left(\dfrac{2x}{1+x}\right)$ implies $x=$ ____

14.(1 pt) Evaluate the limit (Note: enter INF for ∞ and MINF for $-\infty$)

(a) $\lim\limits_{x\to 0^+} x^2e^{-x}=$ ______
(b) $\lim\limits_{x\to 1} x^2e^{-x}=$ ______

15.(1 pt) Evaluate the limit (Note: enter INF for ∞ and MINF for $-\infty$)

(a) $\lim\limits_{x\to 0} e^{-x^3}=$ ______
(b) $\lim\limits_{x\to 1} e^{-x^3}=$ ______

16.(1 pt) Evaluate the limit (Note: enter INF for ∞ and MINF for $-\infty$)

(a) $\lim\limits_{x\to 0^-} e^{1/x}=$ ______
(b) $\lim\limits_{x\to 0^+} \ln(x)=$ ______

17.(1 pt) Evaluate the following expressions.

(a) $\log_2\left(\frac{1}{32}\right)=$ ____
(b) $\log_9 1=$ ____
(c) $\log_4\sqrt{1024}=$ ____
(d) $9^{\log_9 11}=$ ____

18.(1 pt)

$$\ln(r^2s^9\sqrt[10]{r^5s^9})$$

is equal to

$$A\ln r+B\ln s$$

where $A=$ ____ and where $B=$ ____